POLI SCI 403
Probability and Statistics
Fall 2026

Instructor: Gustavo Diaz (gustavo.diaz@northwestern.edu)
Office Hours: By appointment (cal.com/gustavodiaz/)
Office: 2001 Sheridan Rd, suite 3152

Teaching Assistant: Mikhail Turchenko (mturchenko@u.northwestern.edu)
Office Hours: Thursdays 12:00 – 2:00pm
Office: 2001 Sheridan Rd, suite 3169

Lecture: Tuesdays 9:30am – 12:20pm, 2001 Sheridan Rd, suite 3111
Lab: Thursdays 11:00 – 11:50am, 2001 Sheridan Rd, suite 3111
GitHub Repository: github.com/gustavo-diaz/ps403
Canvas: canvas.northwestern.edu/courses/256213

Course Overview

This is the first course in the quantitative methods sequence for graduate students in the Department of Political Science. The course focuses on statistical inference from a social science perspective. Topics include probability, estimation, inference, linear regression, maximum likelihood estimation, identification, and causal inference.

Learning Objectives

The goal of statistical inference is to use the data we have to learn about something for which we do not have data. Establishing that connection requires making assumptions. This course aims to introduce tools and develop skills to conduct statistical inference with as few assumptions as possible.

By the end of the course, you will be able to apply statistical methods to conduct your own analyses, explain statistical tools and concepts in your own words, evaluate the credibility of applied and methodological work, and continue learning more advanced methods.

Prerequisites

There are no formal requirements to take this course other than enrollment in the Political Science PhD program or express approval by the instructor.

This course does not assume prior training on statistics or quantitative methods beyond a grasp of US high school level algebra and calculus (which is covered in math camp). For example, you know that integrals have something to do with calculating the area under a curve, but you do not need to remember how to do integration by hand. I do assume you know how computer file systems work, meaning you are able to determine where a file is located in your laptop and can create new files and folders.

I expect you to participate actively, productively, and respectfully in our meetings. Some of the material addresses complicated concepts or uses math extensively. I do not expect you to understand every single equation for this course, but I do expect you to read carefully enough that you would understand every equation if you chose to revisit the material after taking this course.

Requirements

Reading

The main textbook is:

Textbook

Aronow, P.M. and Benjamin T. Miller. 2019. *Foundations of Agnostic Statistics*. Cambridge University Press

The rest of the syllabus refers to this book as AM. A digital copy is available through the library subscription at no additional cost. You can purchase a physical copy directly from the publisher, although it is usually cheaper on online storefronts. I advise purchasing a physical copy only if you see yourself referencing the book frequently after taking this course.

The book tends to err on the side of brevity and mathematical rigor. Much of our class discussion, assignments, and additional reading will involve untangling and applying the topics in AM. Additional readings should be available through university library subscriptions or distributed promptly. You can find URLs for additional readings in the Schedule section.

Computing

We will use R and RStudio to work on assignments and classroom demonstrations. The advantage of R is that it is free and open source, meaning that you will be able to apply everything you learn in this course anywhere else. The disadvantage is a somewhat steep learning curve. I believe the investment is worthwhile for anyone working with data or in data-adjacent careers. You are welcome to use different software for statistical computing, but I cannot guarantee I will be able to help with troubleshooting.

You can install R and RStudio on your personal computer, which is the preferred use case. You can use this link for installation instructions on Windows and MacOS. See this link for installation instructions on Chromebooks, which is a bit more involved.

You can also use Posit Cloud to access RStudio from any web browser. A free account should suffice for the purposes of this course and has the advantage of letting you access your work across devices.

If you ever need more computing resources than what a personal computer or a free Posit Cloud account allow, you should consider requesting access to the Quest Analytic Nodes from Northwestern IT. I do not anticipate this to be relevant for this course, but it may be useful in the future.

Evaluation

Your final grade in this course will depend on the following:

- Participation
- Lab assignments (8 total, due Fridays 11:59 PM)
- Correspondence paper (due December 9 9:00 AM)

Participation

This course does not formally take attendance, but I do expect the usual level of accountability required in a graduate seminar. That means attending class, doing the reading, asking questions, and working to foster a productive learning environment for everyone. Your participation inside and outside the classroom will be marked as satisfactory or unsatisfactory by the end of the quarter. If your participation leans toward an unsatisfactory mark, I will notify you by the end of Week 6 and give you feedback on how to improve.

Lab assignments

We will have weekly assignments aimed at practicing the application of course material with statistical software. These will range from coding exercises to problems that help illustrate theorems. In general, the goal of the lab assignments is to learn why statistical analyses are (or should not be) conducted in a certain way.

We will dedicate a portion of the weekly lab meeting to lab assignments. This will give you an opportunity to collectively clarify goals and points of confusion. The expectation is that you will come to the meeting having already started working on the lab and have questions ready. On most weeks, you will need to work on your own time to finish the labs. You are welcome to work in groups during our meeting times and beyond, but you must submit individual reports.

Labs are due on Fridays at 11:59PM the week after they are assigned and must be submitted through Canvas in PDF format (you may use the original lab `.qmd` files as templates). Labs will be marked as *satisfactory*, *unsatisfactory*, or *failed* (the last one applies if you did not submit or barely tried). I will mark late submissions as failed unless I give you written approval to submit later. You can resubmit any labs marked as unsatisfactory at any time up until the final paper deadline. If your lab is marked as unsatisfactory, I will give you detailed feedback on what needs to be done to receive a satisfactory mark.

For most labs, I *expect* you to get stuck or be confused. Remember that the purpose of these assignments is to learn, and the quickest path to mastery is making mistakes. I find equal value on getting something right as I do on getting something wrong and doing your best to understand what went wrong or explaining how things should look like if they had gone right.

The implication is that, to assign a satisfactory mark to your lab, I will be mostly looking for evidence that you have learned something useful.

Correspondence paper

You will submit a short correspondence paper as your final assignment. *Correspondence* is a new format adopted by the *American Journal of Political Science* (AJPS) dedicated to scholarship that revisits the central empirical claims in previously published work. In the broader discipline, these are known as replication papers (which is maybe a misnomer).

The goal of the correspondence paper is to apply what you have learned this quarter to a topic of your interest by reproducing the analysis of a previously published article, reflecting on the way it was conducted, consider what could be done differently, and possibly improve upon it.

You should think of this as the first step toward writing a publishable article in your field. Since the AJPS has a dedicated track for this form of scholarship, you are required to choose a paper published in the journal in the last five years (2021-2026).

The paper does not need to be restricted to the methods covered in our class (the only requirement is some form of statistical analysis), but this means you should be willing to work to learn the new methods on your own. I believe this is a valuable skill to practice early on.

You are allowed to work with co-authors or alone as you see fit. I am also open to discussing a different kind of paper or a different target journal if you know what you want to do.

You are required to schedule a meeting with me to discuss the topic of your replication paper. If we do not meet to discuss your paper at least once, your paper will be marked as failed.

Final papers are due on Wednesday, December 9 at 9AM. You should submit your paper in PDF format through Canvas. Your paper should follow the format listed in the AJPS manuscript guidelines, up to 4,000 words and focusing on the main point. You are also required to submit a link to a GitHub repository (or equivalent) including all the code used to reproduce your tables, figures, and calculations. You may also include less important details in an appendix to keep the paper concise. You can read more about the correspondence format here: https://ajps.org/2026/03/05/new-ajps-correspondence-and-corrections-policies/

Your final paper will be marked as *outstanding*, *satisfactory*, *unsatisfactory*, or *failed*. You are also welcome to resubmit a final paper marked as unsatisfactory at any point, but that means I will not be able to update your grade until after the final grade report deadline. I encourage submitting an incomplete paper for a temporary grade over asking for an incomplete course grade.

Grading

This course uses a labor-based grading agreement, commonly known as contract grading. In this course, instead of being given a final grade based on how "good" your submitted assignments are, your final grade will be based on the amount of labor you put into the course. The goal is to decouple grades from performance and emphasize learning and effort.

You will get a default grade if you meet the baseline grading contract. It will go lower if you miss parts of the contract, it will go higher if you meet the baseline plus other criteria.

To meet the baseline grading contract (B+), you should:

- Complete Lab 0
- Be late (by a maximum of 24 hours) on no more than one lab assignment
- Submit the final paper before the deadline
- Complete 6 out of 8 lab assignments with a satisfactory mark
- Receive a satisfactory or outstanding mark in the final paper
- Have a satisfactory participation status by the end of the semester

To get an A-, you should meet the baseline grading contract AND meet at least one of the following:

- Receive a satisfactory mark in 8 out of 8 lab assignments
- Receive an outstanding mark in the final paper

To get an A, you should complete the baseline contract AND both requirements listed to get an A-.

Your grade will go below a B+ if you miss work. Unless otherwise agreed upon in writing on a case-by-case basis, the following criteria outlines how deviating from the baseline contract will impact your grade (assuming everything else constant):

- Participation marked as unsatisfactory: B
- 3 labs marked as unsatisfactory or failed: B-
- 4 labs marked as unsatisfactory or failed: C
- 5 or more labs marked as unsatisfactory or failed: F
- Final paper marked as failed: F

If many of the criteria for missed work apply, you will get the lowest grade applicable.

By signing up for this course, you accept the terms of the grading contract. We will discuss potential amendments in Week 1. Amendments to the grading contract beyond this point should be agreed upon unanimously by students and the instructional team.

Northwestern University Syllabus Standards

This course follows the Northwestern University Syllabus Standards. Students are responsible for familiarizing themselves with this information.

Use of Generative AI Systems

The use of generative artificial intelligence in this course is encouraged as long as it is used to amplify humans instead of replacing them. Any form of cheating, including improper use of content generated by artificial intelligence, constitutes a violation of Northwestern's academic integrity policy. The instructor's general philosophy is that using LLMs for research, statistics, and coding hurts your learning, but can be helpful once you have a solid background.

Copilot is Northwestern University's supported artificial intelligence service. When using Copilot while actively logged in with a Northwestern account, data is stored securely in Northwestern's Microsoft tenant, and Microsoft will not use it for product improvement or to train its AI models.

Schedule

Week 1 (September 29): Preliminaries

Reading:

- AM Introduction

Lab 0: Project-oriented workflow in RStudio

Week 2 (October 6): Probability theory

Reading:

- AM Chapter 1
- Freedman, David A. ad Philip B. Stark. 2003. "What is the chance of an earthquake?" *NATO Science Series IV: Earth and Environmental Sciences*. 32: 201-213
- Spiegelhalter, David. 2024. "Why probability probably doesn't exist (but it is useful to act like it does)." *Nature* 636: 560-564

Lab 1: Calculating and estimating probabilities in R

Week 3 (October 13): Random variables

Reading:

- AM Chapter 2
- Helwig, Nathaniel E. 2020. "Introduction to Random Variables"

Lab 2: Quantities of interest

Week 4 (October 20): Estimation

Reading:

- AM Chapter 3.1-3.3
- Ben Lambert. A full course in econometrics. Videos 6-11

Lab 3: Estimation

Week 5 (October 27): Uncertainty

Reading:

- AM Chapter 3.4-3.6
- Mooney, Christopher. 1996. "Bootstrap Statistical Inference: Examples and Evaluations for Political Science." *American Journal of Political Science* 40 (2): 570-602
- Gelman, Andrew. 2023. "What is a standard error?" *Journal of Econometrics* 237 (1): 105516
- Wooldridge, Jeffrey M. 2023. "What is a standard error? (And how should we compute it?)" *Journal of Econometrics* 237 (1): 105517 (Section 1 only)
- Powell, James L. 2023. "Discussion of 'What is a standard error?'" *Journal of Econometrics* 237 (1): 105518

Lab 4: Quantifying uncertainty

Week 6 (November 3): Regression

Reading:

- AM Chapter 4
- Berk, Richard. 2010. "What You Can and Can't Properly do with Regression." *Journal of Quantitative Criminology* 26: 481-487

Lab 5: Adjusting for covariates

Week 7 (November 10): Parametric models

Reading:

- AM Chapter 5
- James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2021. *An Introduction to Statistical Learning with Applications in R*. Springer. Chapters 4.1-4.3
- Achen, Christopher. 2002. "Toward a New Political Methodology: Microfoundations and ART." *Annual Review of Political Science* 5: 423-450

Lab 6: Maximum likelihood estimation

Week 8 (November 17): Missing data

Reading:

- AM Chapter 6
- Lall, Ranjit. 2016. "How Multiple Imputation Makes a Difference." *Political Analysis* 24 (4): 414-433
- Arel-Bundock, Vincent and Krzysztof J. Pelc. 2018. "When Can Multiple Imputation Improve Regression Estimates?" *Political Analysis* 26 (2): 240-245
- Pepinsky, Thomas. 2018. "A Note on Listwise Deletion versus Multiple Imputation." *Political Analysis* 26 (4): 480-488

Lab 7: Dealing with missing data

Week 9 (November 24): Causal inference

Reading:

- AM Chapter 7
- Keele, Luke, Corrine McConnaughy, and Ismail White. 2012. "Strengthening the Experimenter's Toolbox: Statistical Estimation of Internal Validity." *American Journal of Political Science* 56 (2): 484-499

Lab 8: Randomization inference and hypothesis testing

Week 10 (December 1): Review/make-up class